ways was responsible for all of chemistry. A simple summary, but one with world-altering implications.

Dalton jumped the gun a little bit with his nomenclature. For the Greeks, the whole point of atoms was that they were indivisible, the fundamental building blocks out of which everything else is made. But Dalton's atoms are not at all indivisible—they consist of a compact nucleus surrounded by orbiting electrons. It took over a hundred years to realize that, however. First the English physicist J. J. Thomson discovered electrons in 1897. These seemed to be an utterly new kind of particle, electrically charged and only 1/1800th the mass of hydrogen, the lightest atom. In 1909 Thomson's former student Ernest Rutherford, a New Zealand physicist who had moved to the UK for his advanced studies, showed most of the mass of the atom was concentrated in a central nucleus, while the atom's overall size was set by the orbits of much lighter electrons traveling around that nucleus. The standard cartoon picture of an atom, with electrons circling the nucleus much like planets orbit the sun in our solar system, represents this Rutherford model of atomic structure. (Rutherford didn't know about quantum mechanics, so this cartoon deviates from reality in significant ways, as we shall see.)

Further work, initiated by Rutherford and followed up by others, revealed that nuclei themselves aren't elementary, but consist of positively charged protons and uncharged neutrons. The electric charges of electrons and protons are equal in magnitude but opposite in sign, so an atom with an equal number of each (and however many neutrons you like) will be electrically neutral. It wasn't until the 1960s and '70s that physicists established that protons and neutrons are also made of smaller particles, called quarks, held together by new force-carrying particles called gluons.